

NativeX EdTech Group: Using Big Data to recommend custom curriculum for English- language learning

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ABOUT NATIVEX



About NativeX

With Google Cloud, NativeX EdTech Group enables Vietnamese people to improve their career prospects and access to higher education by learning English via featured online programs.

NativeX EdTech Group is an education technology (edTech) company that focuses on improving career prospects and access to higher education for 18-35 year-olds in Vietnam and other developing markets. Since its foundation, the company has grown to more than 180 employees and 300-plus teachers, based primarily in Europe.

Industries: Education

Google Cloud results

- Reduces the number of team members needed to administer platform infrastructure by 85%
- Enables teams to focus on product development and customer experience by eliminating an hour a day spent scaling resources
- Gained the ability to support 1,000 concurrent classes, if required, without incurring

Location: Vietnam

Reduced
infrastructure
costs than
running
educational
technical
platform
scaling
environment

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud (<https://cloud.google.com/>)

(<https://cloud.google.com/products/operations>)

BigQuery (<https://cloud.google.com/bigquery/>)

(<https://workspace.google.com/>) Google

Cloud SQL (<https://cloud.google.com/sql/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Firebase (<https://firebase.google.com/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Operations

(<https://cloud.google.com/products/operations>)

Google Workspace

(<https://workspace.google.com/>)

Google Drive

(<https://workspace.google.com/products/drive/>)

**significant
additional
costs**

In Vietnam, being proficient in the English language can be the gateway to opportunities in travel and tourism, businesses with international operations, and higher education. Unfortunately, the fast-developing South East Asian country still lacks offline courses and centers to meet the demand for English-language training. For a country that aims to reach high-income status by 2045

(<https://www.worldbank.org/en/news/press-release/2022/03/25/vietnam-world-bank-strengthen-partnership-to-meet-goal-of-high-income-economy-by-2045>)

, educating more of its people in the international language of business is an important developmental step.

This is where NativeX EdTech Group

(<https://nativex.edu.vn/>) and its fully-featured education platform running on Google Cloud come in.

NativeX EdTech Group enables adults in Vietnam to learn English online through its platform, which features comprehensive learning content, including lessons from international teachers and study programs. With offices in Ho Chi Minh City and Hanoi, NativeX EdTech Group also works closely with platform users to determine the best pathway for them to learn English.

Disparate platforms made unified analysis difficult

NativeX EdTech Group was previously running several independently-developed systems managed in isolation by small teams. Each system would deliver an individual platform component or feature, such as class delivery or vocabulary testing.

Unfortunately, with these systems running in different clouds run by local or multinational providers, the business found it difficult to provide a unified analysis of user behavior as well as challenges to grow in other countries.

"We had to undertake extensive data pipeline work across multiple environments to aggregate, consolidate and analyze data," says Sandeep Jain, CTO at NativeX EdTech Group.

The business also had to dedicate many separate infrastructure teams to administer its various cloud environments, including local services in South East Asian markets that sometimes fell short of delivering the required experience when connecting teachers based primarily in Europe with students in Vietnam. "When we run live classes, we want the system to load quickly and stream video of classes without delay while capturing all interactions and activities," says Jain.

Doing so imposes a significant load on the NativeX EdTech Group infrastructure. The business runs 10,000 to 12,000 classes per week with around three to six students per class depending on their subscription type, with load peaking at around eight

to nine o'clock on weekday evenings. To deliver on these requirements, the business's technology team had to spend an hour a day manually adding and removing resources to its infrastructure to match demand peaks and troughs.

Automation, scaling, and managed services ease cloud decision

"We decided to migrate our education platform to Google Cloud, with its automation, managed services and seamless scalability driving our decision," says Jain. The business also wanted a service that could connect seamlessly to and complement a MongoDB database used to store all users' learning activity data to analyze learning behavioral patterns.

Jain and the NativeX EdTech Group development team found Google Cloud easy to launch and use, and able to adapt for the team even without prior experience, which was a significant technical advantage over competing cloud platforms. "The fact that most of the core services required by Google Cloud were fully managed, and we could easily set up and run data pipelines on the platform, with data transferred seamlessly to a data warehouse for analysis, reinforced our decision," says Jain.

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*—Sandeep Jain, CTO at
NativeX EdTech Group*

NativeX EdTech Group worked closely with the Google Cloud team and partner [Cloud Ace](https://www.cloud-ace.com/) (https://www.cloud-ace.com/) to implement a consolidated cloud architecture. Experts who worked on the business's use case recommended an architecture comprising a [BigQuery](https://cloud.google.com/bigquery) (https://cloud.google.com/bigquery) analytics data warehouse, [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (https://cloud.google.com/kubernetes-engine) to provide a managed Kubernetes platform to automate the

deployment and scaling of containerized applications, and Operations (<https://cloud.google.com/products/operations?hl=en>) to enable the business to monitor and troubleshoot its infrastructure, software, and applications.

With the NativeX EdTech Group platform running as a microservices architecture in Google Kubernetes Engine, the business has automated the process of increasing or decreasing resources in line with demand. "We no longer need to dedicate time or resources for managing systems like before, which frees our team to spend more time on product development and customer experience," says Jain.

Recommending appropriate learning paths to improve English

With BigQuery, NativeX EdTech Group is analyzing the learning behavior of its users and the performance reports that detail the extent to which users are improving their skills and test results. This enables the organization to recommend appropriate learning paths that give users the best opportunity possible to improve their English language capabilities. "That helps the business in a big way. For example, we can advise students which skills to improve or what topics they should retake," says Jain. "We couldn't make these recommendations before, but doing so now means we can improve our product." Analyzing data through the cloud data warehouse also enables the business to increase user registrations and conversions, as the company was able to acquire the same number of users with

just 60% of its marketing budget after moving to Google Cloud.

The Firestore (<https://firebase.corp.google.com/>) mobile platform enables the organization to undertake a range of business-critical tasks, including user authentication, optimizing marketing campaign performance through A/B testing, and configuring features of its platform remotely. In particular, the business can deliver push notifications to mobile users to advise them, for example, that they have a class starting in 30 minutes, without incurring the costs of doing so through SMS messages.

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our content
writers or
teachers use
Drive to store,
access and
share what
they're working
on. When that
process is
complete, they
upload it to our
platform for use
in our English
language
learning
products."

*—Sandeep Jain, CTO at
NativeX EdTech Group*

NativeX EdTech Group uses Cloud SQL
(<https://cloud.google.com/sql>) to store user
authentication and payment transaction data in a
relational database running in a secure, scalable,
and fully integrated environment, while Cloud
Storage (<https://cloud.google.com/storage>) securely
stores a wide range of media, such as video files,

audio files, and documents in an easily accessible repository.

Drive (<https://workspace.google.com/products/drive/>) also plays a key role in the NativeX EdTech Group platform. "We need to create a lot of online content in-house derived from our partnerships with large educational resource providers such as National Geographic Learning and Macmillan Education, so our content writers or teachers use Drive to store, access and share what they're working on," says Jain. "When that process is complete, they upload it to our platform for use in our English language learning products."

Reduces infrastructure costs

Moving to Google Cloud initially enabled NativeX EdTech Group to reduce its infrastructure costs by 70%, largely by eliminating the need to manually provision additional servers to meet demand peaks. However, as its internal teams optimized running the platform within Google Kubernetes Engine, this figure has risen to 80%, enabling the business to invest more in marketing and product development.

The business has also reduced the size of its infrastructure management team by approximately 85%, with those team members combining their few management tasks with development work. Furthermore, cloud and data infrastructure availability of 99.99% means NativeX EdTech Group delivers an experience that retains existing users and builds its reputation among prospective English learners.

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Ease of use is another key benefit. "Google Cloud makes DevOps jobs easy by investing and developing a system that is intuitive and straightforward," says Jain. "If I want to undertake a launch, I can do it very quickly without concerning myself with setup."

With Google Cloud, NativeX is positioned to support growth. "With Google Cloud, I can easily support more traffic. If I want to run 1,000 classes in parallel, I can do so without incurring a cost impact," concludes Jain. "We can scale and grow quickly, and realize our ambition of helping Vietnam's younger generation join the global economy with confidence thanks to proficiency in English."